Chapter test

12 The cell cycle and cell division

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

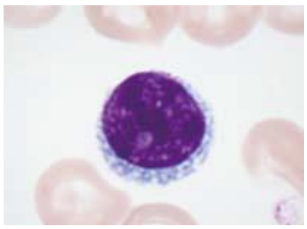
A. Concept checking (10 marks, 1 mark each)

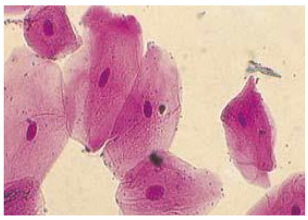
1. The body cells of horses contain 64 chromosomes. How many chromosomes are there in a horse gamete?

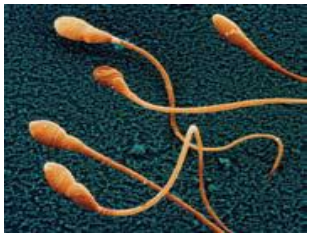
   A. 16
   B. 32
   C. 64
   D. 128

★ 2. The photomicrographs below show four types of human cells.

(1) 

(2) 

(3) 

(4) 

Which of the following cell types are diploid?

A. (1) and (2) only
B. (2) and (3) only
C. (1), (2) and (3) only
D. (1), (2), (3) and (4)

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3. Which of the following statements about cell divisions are correct?

(1) The parent cell gives rise to two daughter cells in mitosis.
(2) Chromosomes replicate once in mitosis and twice in meiosis.
(3) The daughter cells formed in meiosis have the same number of chromosomes.

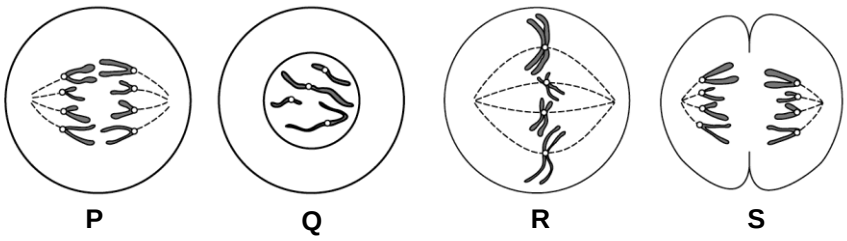
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

★ 4. Which of the following events take(s) place during interphase?

(1) DNA replication
(2) breakdown of nuclear membrane
(3) synthesis of mitochondria

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

5. Below are four different stages of mitotic cell division.



Which of the following shows the correct sequence?

A. Q → R → P → S
B. P → R → S → Q
C. R → P → Q → S
D. Q → P → R → S ☐

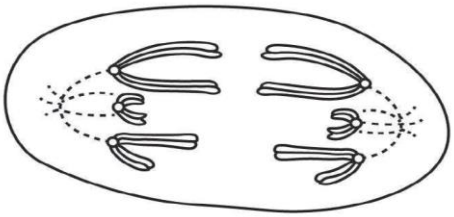
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★ 6. Which of the following is/are associated with mitosis?

(1) repair of damaged tissues
(2) growth of tumors
(3) development of a zygote

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

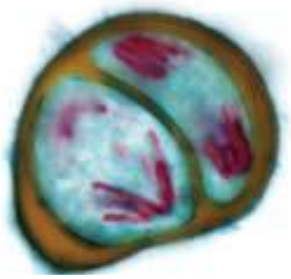
★ 7. The diagram below shows a cell undergoing meiosis.



Which stage of meiosis is shown in the diagram?

A. metaphase I
B. metaphase II
C. anaphase I
D. anaphase II ☐

★★8. The photomicrograph below shows a cell undergoing cell division.



This type of cell division occurs

A. in the skin.
B. in the testis.
C. in the anther.
D. at the root tip. ☐

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★ 9. The cancer drug Vinblastine interferes with the synthesis of spindle fibres. Which stage of the cell cycle of cancer cells will be hampered by this drug?

A. interphase
B. prophase
C. anaphase
D. telophase ☐

10. Genetic variation in gametes is brought about by

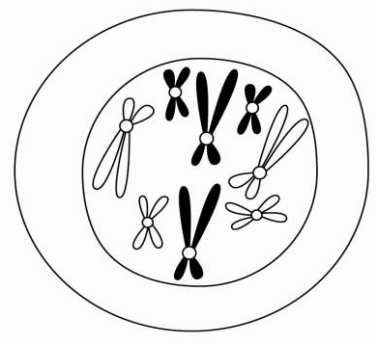
(1) crossing over of homologous chromosomes.
(2) independent assortment of chromosomes.
(3) random fertilization.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	12.1	12.1	12.2	12.2	12.2	12.2	12.3	12.3	12.3	12.3

B. Problem solving (19 marks)

11. The diagram below shows a cell carrying out mitosis.



With reference to the above diagram, state the number of each of the following.

(a) chromosomes (1 mark)

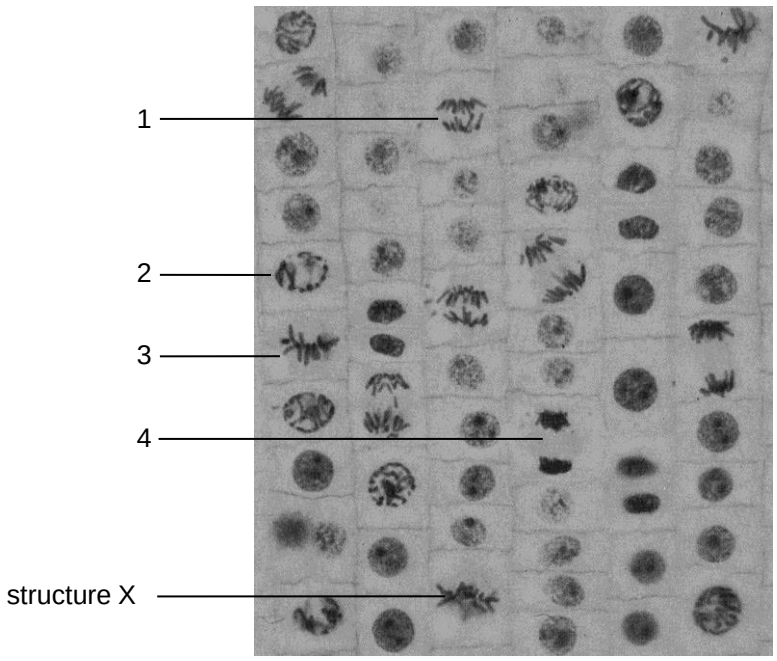
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(b) chromatids (1 mark)

(c) pairs of homologous chromosomes (1 mark)

(Total: 3 marks)

12. The photomicrograph below shows a section of an onion root tip which presents the different stages of cell division.



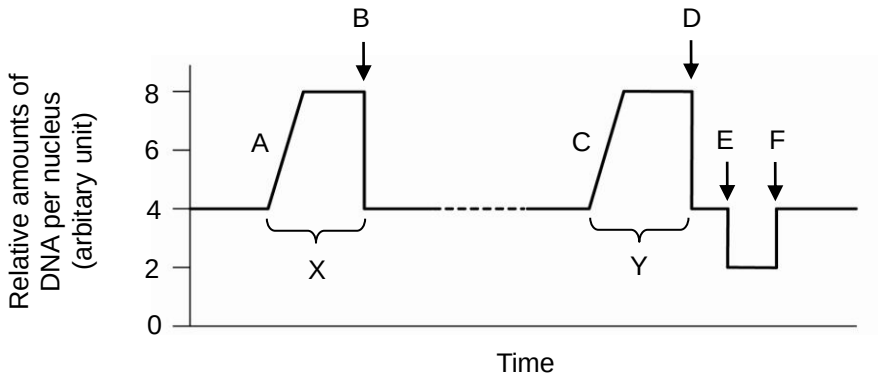
(a) Name structure X. What is the importance of structure X in a cell? (3 marks)

(b) Arrange the cells labelled 1 to 4 in the correct sequence of cell division. (1 mark)

(Total: 4 marks)

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★ 13. The graph below shows the change in the relative amounts of DNA per nucleus during two cell divisions in an animal tissue.



(a) Name the types of nuclear divisions X and Y. Explain your answer. (4 marks)

(b) Deduce what is happening at points A, B and C. (4 marks)

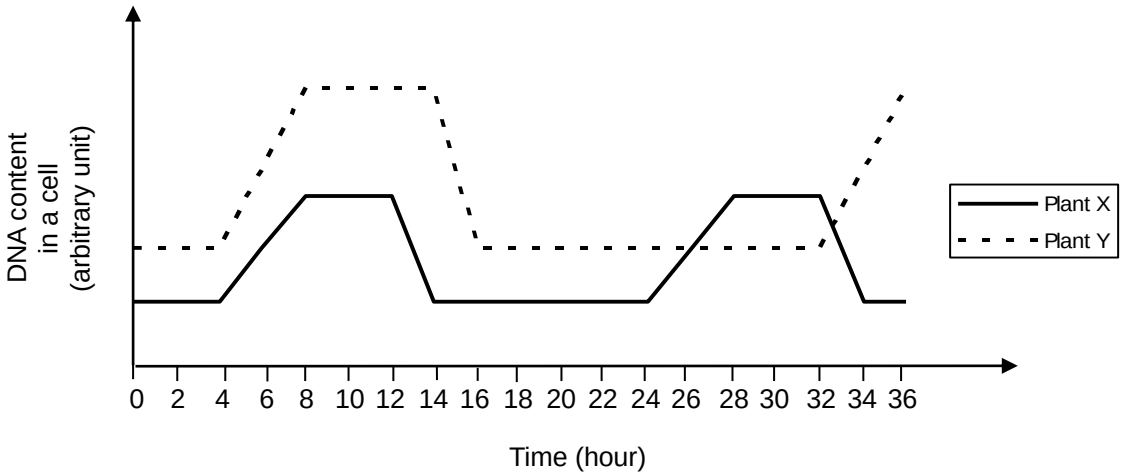
(c) Deduce what is happening at points E and F. (4 marks)

(Total: 12 marks)

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C. Exam challenge (11 marks)

★★14. A scientist studied the cell cycle of two different plants, X and Y. The graph below shows the changes in the DNA content of the plant cells.

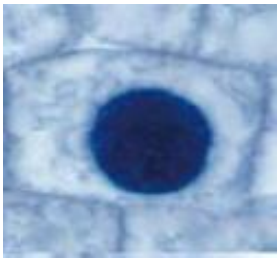
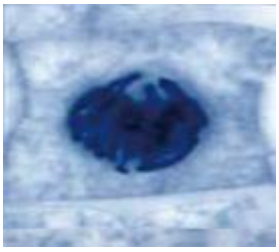
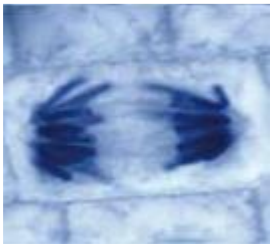


(a) Compare the cell cycle of plants X and Y. (2 marks)

(b) In plant Y, the DNA content of the cells doubles between the 4th and 8th hour. Explain this change in DNA content and its significance. (2 marks)

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(c) The scientist further examined 300 cells of plant X and counted the number of cells in different stages of the cycle. The table below shows some of the cells he observed, and the percentage of cells in interphase and in two stages of mitosis, A and B.

Stage of cell cycle	Photomicrograph	Percentage of cells (%)
Interphase		92
Stage A		2
Stage B		1

(i) Explain why the scientist need to examine 300 cells instead of fewer cells. (1 mark)

(ii) Identify stages A and B. Give the evidences from the photomicrographs to support your answer. (4 marks)

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(iii) With reference to the graph, calculate the relative length of stage A of the cell cycle of plant X. (2 marks)

(Total: 11 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	B	2.	B	3.	B	4.	B	5.	A
6.	D	7.	C	8.	C	9.	C	10.	A

(1 mark each)

B. Problem solving (19 marks)

11. (a) 8 (1)
    (b) 16 (1)
    (c) 4 (1)

(Total: 3 marks)

12. (a) Structure X is the chromosome. (1)
    Chromosomes contain DNA (1)
    which store genetic information for making proteins for growth and functioning. (1)
    (b) 2 → 3 → 1 → 4 (1)

(Total: 4 marks)

13. (a) X is mitosis (1)
    because at the end of the division the amount of DNA per nucleus remains unchanged. (1)
    Y is meiosis (1)
    because at the end of the division the amount of DNA per nucleus reduces by half. (1)
    (b) At A and C, DNA molecules are undergoing replication. (1)
    Therefore, the amount of DNA doubles at A and C. (1)
    At B, the amount of DNA halves to become normal. (1)
    This is due to the division of nucleus into two daughter nuclei. (1)
    (c) At E, meiosis II occurs. (1)
    As a result, each daughter nucleus has half of the amount of DNA. (1)
    At F, fertilization occurs. (1)
    Two haploid gametes fuse to produce a diploid zygote. (1)

(Total: 12 marks)

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C. Exam challenge (11 marks)

14. (a) Any ***two*** of the following: (2)
    - The cell cycle is shorter in X than in Y. (1)
    - Interphase is shorter in X than in Y. (1)
    - Mitosis / the nuclear division is shorter in X than in Y. (1)
    - The time required for DNA replication is the same in both plants. (1)

    (b) The cells are preparing to divide. (1)
    DNA is replicated so that the new cells will have same quantity of DNA as the parent cell after division. (1)

    (c) (i) To ensure the percentage recorded is more accurate. (1)

    (ii) Stage A is prophase. (1)
    The chromosomes in the nucleus become visible. (1)
    Stage B is anaphase. (1)
    The chromosomes are separating. (1)

    (iii) 20 x 2% (1)
    = 0.4 hour (1)

(Total: 11 marks)

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